

MICHAEL B. GREENBAUM

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SUMMARY

Software Engineer with 2 years of C++ and Python experience building scalable systems at Lockheed Martin. Published first author in IEEE BigData on GPU-accelerated algorithms, achieving 3,992x speedup on an NP-hard problem. Graduated summa cum laude from Rowan University.

EDUCATION

Rowan University, Glassboro, NJ **Graduated May 2026**

Bachelor of Science in Computer Science

Minor in Mathematics, Honors Concentration, GPA 3.9/4.0

Awards: Summa Cum Laude, CS Outstanding Senior Award, *YITE* CS Honors Society, *TIME* Math Honors Society, President's List for 4 Semesters, Dean's List for 2 Semesters, Rowan University Scholars Scholarship

Relevant Coursework: Object-Oriented Programming, Data Structures and Algorithms, Software Engineering, Computer Org., Database Systems, Calculus I–III, Discrete Math, Linear Algebra, Probability and Random Var., Statistical Analysis

EXPERIENCE

Digital Simulations Team, Lockheed Martin Corporation, Camden, NJ **January 2025–Present**

Software Engineer; Associate Member Engineering Staff (E1)

- Engineered a low-fidelity component factory enabling rapid prototyping of simulation components with variable parameters. Greatly accelerated early-stage testing by bypassing the overhead of the high-fidelity main simulation. To keep the factory low-effort, I developed a multitude of workarounds; debugging them became the main blocker to integrating the system.
- Refactored the simulation's communication system into a scalable message processor. Significantly reduced the maintenance required to add new messages by planning for growth. I achieved this extensibility by implementing a queue-based buffer that leveraged generic types. Authored user guides and maintenance documentation to support team adoption.
- Architected a loosely coupled external interface to bridge the simulation with an external component. Enabled customer-specific software deployments by making the interface easily removable. I heavily abstracted and encapsulated the component by wrapping its functionality into new operations. This allows the system to utilize the component's functionality without needing to understand its inner workings.

High-Performance Data Mining Lab, Rowan University, Glassboro, NJ **October 2022–January 2025**

Project Manager, "GPU-Based Graph Mining"; Research Assistant

- Led a multi-year research project to develop the first GPU-accelerated algorithm for the NP-Hard maximal γ -quasi-clique problem. Programmed an algorithm 3,992x faster than the CPU baseline; the related paper was published in a top conference. We invented novel data structures and processes to enable the transition to the GPU. We further optimized the algorithm by leveraging parallel distributed techniques, specifically a three-way handshake approach that uses up to 4 GPUs simultaneously.

ACTIVITIES

First Author of Published Research: Michael Greenbaum, Ke Fan, Wajid Manzoor, and Guimu Guo. "GPU-Accelerated Maximal Quasi-Clique Mining." Dec 8, 2025. In 2025 IEEE International Conference on Big Data (BigData), pp. 152-161.

International Collegiate Programming Competition: Mid-Atlantic USA 2024, Division 2, First Place

LeetCode: Global Rank: 1,796,088 (Top 11.97%), Contest Rank: 176,796 (Top 20.31%)

SKILLS

Programming Languages: Python, Java, C/C++, CUDA C++, MPI, Pthreads, OpenMP, Makefile, Bash Scripting

OS, IDE, and Applications: Windows, Linux, Visual Studio, VIM, GitHub, Atlassian Suite, Git, GDB, GCC

Security Clearance: Secret